

AS Chemistry — Topic 1 Atomic Structure (Exam)

Teacher mark scheme · CIE 9701 · 34 marks

This file is for teachers only. CIE IDs and letters are listed so the paper can be remapped. Student **main.pdf** has questions only.

Section A — MCQ key

Q	Letter	CIE ID	Paper
A1	A	cie-9701-2021-fm-p12-q1	2021 FM P12 Q1
A2	B	cie-9701-2025-fm-p12-q4	2025 FM P12 Q4
A3	A	cie-9701-2024-mj-p13-q3	2024 MJ P13 Q3
A4	C	cie-9701-2025-mj-p12-q14	2025 MJ P12 Q14
A5	D	cie-9701-2025-on-p11-q40	2025 ON P11 Q40
A6	A	cie-9701-2025-mj-p13-q1	2025 MJ P13 Q1
A7	D	cie-9701-2025-on-p12-q1	2025 ON P12 Q1
A8	B	cie-9701-2021-on-p12-q4	2021 ON P12 Q4
A9	C	cie-9701-2021-mj-p11-q7	2021 MJ P11 Q7
A10	C	cie-9701-2022-mj-p12-q1	2022 MJ P12 Q1
A11	C	cie-9701-2025-mj-p11-q4	2025 MJ P11 Q4
A12	D	cie-9701-2023-on-p11-q2	2023 ON P11 Q2
A13	B	cie-9701-2021-mj-p13-q31	2021 MJ P13 Q31

B1 · 2025 MJ Paper 21 Question 2 (11 marks)

- 1 atoms (of an element) with same number of protons and different number of neutrons cie-9701-2025-mj-p21-q2a
- 2 M1 (5.9×54), (91.9×56), (2.2×57). M2 $A_r = 55.9$ (1 d.p.) cie-9701-2025-mj-p21-q2b-i
- (ii) [2] protons **26**; nucleons **56** cie-9701-2025-mj-p21-q2b-ii
- 1 **5** cie-9701-2025-mj-p21-q2c
- 1 $\text{Fe(g)} \longrightarrow \text{Fe}^+(\text{g}) + \text{e}^-$ cie-9701-2025-mj-p21-q2d
- 4 M1 same electronic configuration / same shielding. M2 same number of protons / same nuclear charge. M3 same nuclear attraction to outer-shell electron. M4 no change / will not be affected. cie-9701-2025-mj-p21-q2e

B2 · 2025 MJ Paper 23 Question 1 (10 marks)

- 1 $1s^2 2s^2 2p^5$ cie-9701-2025-mj-p23-q1a-i
- (ii) [1] **2** pairs cie-9701-2025-mj-p23-q1a-ii
- (iii) [1] spherical s orbital (circle) cie-9701-2025-mj-p23-q1a-iii
- 1 $\text{S(g)} \longrightarrow \text{S}^+(\text{g}) + \text{e}^-$ cie-9701-2025-mj-p23-q1b-i
- (ii) [2] M1 pair of electrons in a 3p orbital (in S) repel. M2 outweighs increase in nuclear charge. cie-9701-2025-mj-p23-q1b-ii
- (iii) [4] M1 $\text{Na}^+ < \text{Ne} < \text{F}^-$. M2 isoelectronic / same number of electrons. M3 larger species has smaller nuclear charge / fewer protons. M4 smaller nuclear attraction. cie-9701-2025-mj-p23-q1b-iii

Marks by learning outcome

LO cluster	Marks
1.1 particles / nucleon number / electric field / radius	9
1.2 isotopes	2
1.2 / 2.1 / 22.2 A_r from abundances / mass spectrum	3
1.3 configuration / orbitals / unpaired electrons	7
1.4 ionisation energy	13
Total	34

Quick MCQ string: **A B A C D · A D B C · C C D B**Full exclusion check: `../TEACHER-KEY.md`.